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RESEARCH ARTICLE

UV Spectrophotometric Analytical Method Development and Validation for Aloe-Emodin Phytochemical in Bulk and Formulations

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ABSTRACT:

Aloe-emodin is an anthraquinone derivative mainly obtained from *Aloe vera* plant. It is reported to have several pharmacological activities including antiviral, antiproliferative, analgesic, antidiarrheal etc. Several formulations are available in market in combination with other phytoconstituents. Hence, there is a need for an analytical method for regular analysis of aloe-emodin in marketed formulations and bulk drugs. The goal of the current study was to use a UV 1800 Shimadzu double beam spectrophotometer to design and validate a straightforward, quick, accurate, and selective analytical method for estimating aloe-emodin in bulk and in marketed formulations. Using methanol as the solvent, the spectrophotometer-based detection process was carried out at λ_{max} of 428.5 nm. The linearity, range, specificity, accuracy/recovery, precision, and robustness of the suggested approach were all confirmed. With a correlation coefficient of more than 0.999, the aloe-emodin detector response was found to be linear over the chosen concentration range (0.1-0.7 $\mu\text{g/mL}$). The accuracy was between 98.78 to 100.38%, variance of standard curve line was found to be 0.052. Among the five sample preparations, the precision (R.S.D.) ranged from 0.129 to 0.325%. The limits of quantification (LOQ) and detection (LOD) were reported to be 26.518 $\mu\text{g/mL}$ and 0.278 $\mu\text{g/mL}$, respectively. The recovery of aloe-emodin was about 98.78 to 100.38%. The data collected showed that the commercial formulations' excipients did not interfere with the procedure. As a result, it is convenient to use these formulations for routine quality control analyses of aloe-emodin in bulk medication, marketed capsule, and other formulations.

KEYWORDS: *Aloe vera*, Phytochemical, UV1800, Analytical method, Development, Validation.

INTRODUCTION:

Aloe vera, *Cassia occidentalis*, *Rheum palmatum* L., *Polygonum multiflorum* Thunb, and other commonly used Chinese medicinal herbs are sources of aloe-emodin, a naturally occurring anthraquinone derivative that is chemically 1,8-dihydroxy-3-hydroxymethyl anthraquinone (Figure 1).

These herbs are used as traditional medicines in many different nations.¹ Aloe-emodin is currently receiving a lot of interest because to its exceptional antineoplastic effect on a variety of tumor cells, such as those from the stomach, liver, lung, melanoma, breast, and colon cancers.²⁻⁴ Additionally, it has been shown that aloe-emodin possesses a broad range of pharmacological properties, including antiviral, anti-inflammatory, antibacterial, antiparasitic, neuroprotective, and hepatoprotective activities.⁵⁻⁷ It has been used to treat a variety of conditions, including the influenza virus, inflammation, sepsis, Alzheimer's disease, glaucoma,

malaria, liver fibrosis, Type 2 diabetes, growth issues, and several cancers.

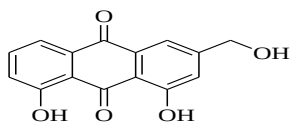


Figure 1: Chemical structure of aloë-emodin (1,8-dihydroxy-3-hydroxymethyl-anthraquinone)

Lang (1993) were the first to report that oral administration of ^{14}C aloë-emodin (4.5mg/kg) was quickly metabolized to rhein and other unknown metabolites with a terminal half-life in blood of 50 h then ultimately excreted in urine and feces.⁸ After intraperitoneal and intragastric administration of aloë-emodin to rats, it was demonstrated that aloë-emodin was excreted by first-order kinetics and that the concentration-time curve of aloë-emodin fit the two-compartment model.⁹⁻¹⁰ Literature survey revealed that a variety of analytical methods viz. LC, HPLC, HPLC/FLD, UPLC, LC/MS, GC/MS, TLC and HPTLC has been developed for the analysis but as simultaneous estimation method in plasma and urine or in formulations.¹¹⁻¹³ It is necessary to establish an analytical approach that is straightforward, sensitive, quick, and accurate for estimating aloë-emodin in pure form and in pharmaceutical preparations because the formulations are available without combinations of any medications.¹⁴⁻¹⁵

Consequently, the current research work's objective is to create a validate data for UV-Visible spectrophotometer analysis method that can be simply implemented as a regular in quality testing laboratories. This has allowed us to lower the overall analysis time while also addressing the inaccuracy brought on by insufficient extraction and the use of internal standards.¹⁶⁻¹⁹

MATERIALS AND METHODS:

Materials:

Aloë-emodin phytochemical from Avra Synthesis Pvt Ltd Ananthpuram, Andhra Pradesh; Aloë-emodin powder for dietary supplement and aloë-emodin capsule, label claim equivalent to 500mg, procured from Botany and Alfital respectively; UV1800, Shimadzu, with 10mm matched quartz cells; analytical-grade methanol, Rankem Chemicals.

Method development:

Solvent selection and optimization:

Peak quality and lack of interference at the designated wavelength were investigated for chloroform, acetone, methanol, of which, methanol met all requirements.²⁰

Preparation of stock solution and standard serial dilutions and selection of the analytical wavelength:

Stock solution (10 $\mu\text{g/mL}$) of aloë-emodin in methanol were prepared, and aliquots were divided into fractions and transferred to a series of 10mL volumetric flasks to produce standard dilutions from 0.1 to 0.7 $\mu\text{g/mL}$. A 0.4 $\mu\text{g/mL}$ standard solution of aloë-emodin was scanned between 800 to 200nm on UV 1800 for the wavelength at which appreciable absorption without much noise took place.

Validation of the method:

The method was evaluated and analyzed for linearity, range, sensitivity, specificity and selectivity, accuracy/recovery, intraday and inter day precision and robustness as per the International Conference on Harmonization (ICH) Q2 (R1) guidelines.²¹

Linearity and range:

Seven different standard dilutions of aloë-emodin (concentrations range 0.1-0.7 $\mu\text{g/mL}$) were prepared in methanol from a fresh stock solution of 10 $\mu\text{g/mL}$ and were analyzed quantitatively at λ_{max} 428.5nm. The obtained data was subjected to least square regression analysis in order to determine linearity and range.²²

Sensitivity (LOD and LOQ):

known concentrations (0.1-0.7 $\mu\text{g/mL}$) of aloë-emodin were diluted until the average responses for five replicate measurements were roughly three to ten times the standard deviation of the answers. LOD and LOQ were computed using the following formulas:

$$\text{LOD} = 3.3 \sigma/S \dots\dots\dots (i)$$

$$\text{LOQ} = 10 \sigma/S \dots\dots\dots (ii)$$

Specificity and selectivity:

Three separate solutions of 0.4 $\mu\text{g/mL}$ concentrations from individual stock solutions were created for the marketed formulations. Standard aloë-emodin, produced in methanol, was then examined and compared with a pure aloë-emodin solution of the same strength.²³

Accuracy/recovery:

Three distinct stock solutions were used to generate three different concentrations of the standard aloë-emodin in methanol (0.3, 0.4, and 0.5 $\mu\text{g/mL}$), and the standard curve was used to estimate the strengths of each concentration independently by adding three distinct standard concentrations of aloë-emodin (0.05, 0.10, and 0.15 $\mu\text{g/mL}$) to the previously analyzed aloë-emodin solutions and analyzing them again in the same manner. Based on actual and estimated concentrations, the accuracy was presented as % recovery (% confidence interval) with % relative error.²⁴

Intraday and inter day precision (repeatability and reproducibility):

Repeatability and reproducibility were validated by examining three distinct aloe-emodin concentrations (0.3, 0.4 and 0.5 µg/mL) created in methanol in pantalets on the same day and on different days. For each set of data, the estimated concentrations based on the standard curve were given a confidence range, standard deviation, and percentage relative standard deviation.²⁵

Robustness:

The proposed method's robustness was examined by doing tests on pure aloe-emodin and the medicine in capsule form while slightly adjusting the analysis's maximum wavelength (428.5nm) by ± 1.0 nm. %relative error, percentage of mean recovery and confidence interval with 95% confidence range were determined and reported.²⁶

Application of the suggested methodology to the commercial formulations' assay:

The extraction of the aloe-emodin, emptying twenty capsules and removing a piece of the powder equal to 10 mg drug from the powder excipients was executed in methanol by doing sonication for 30 minutes and making the final volume to 100mL. The resultant mixture was centrifuged at 2500 rpm for 10 minutes, and the supernatant's drug content was examined. Three independent stocks of aloe-emodin in methanol at a 10 µg/mL concentration were made using two different aloe-emodin capsule formulations that are commercially available. To create three distinct concentrations of aloe-emodin (0.3, 0.4, and 0.5 µg/mL), these three stocks were each used separately. The suggested procedure was used to evaluate the prepared solutions and the assay data are presented along with the confidence intervals.²⁷

RESULTS AND DISCUSSION:

The proposed approach was shown to be straightforward, sensitive, accurate, precise, inexpensive, and quick. A spectral scan revealed that the peak of aloe-emodin was at 428.5nm (Figure 2a).

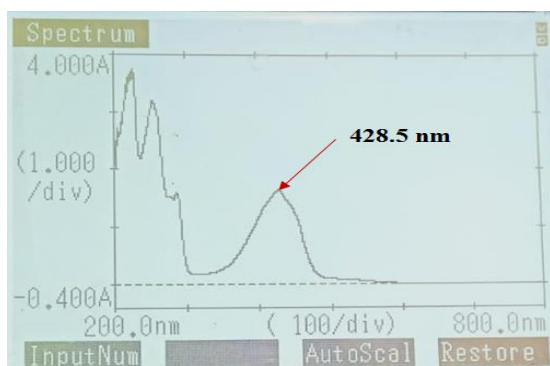


Figure 2: a) Scanning of aloe-emodin in methanol for absorption maxima (λ_{max} =428.5nm);

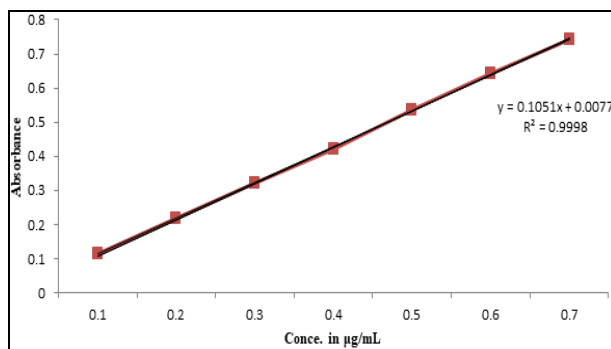


Figure 2: b) Calibration curve of aloe-emodin in methanol at 428.5 nm

The method was validated for as per ICH guidelines (Q2 (R1) (ICHHT, 2022). A good linear relationship between absorbance and concentration was found in the chosen concentration range (0.1-0.7 µg/mL). Characteristic parameters are linear dynamic range 0.1-0.7 µg/mL, regression equation $y=0.105x+0.007$, slope, correlation coefficient 0.999, SE intercept 0.004, SD intercept 0.011, LOD (µg/mL) 0.278, LOQ (µg/mL) 26.518 (Figure 2b and Table 1).

Table 1: Regression and optical features of the suggested technique summarized

Parameters	Values
Linear dynamic range (µg/mL)	0.1-0.7
Regression equation	$y = 0.1051x + 0.0077$
Correlation coefficient (r)	$R^2 = 0.999$
SE intercept	0.00418
SD intercept	0.01105
LOD (µg/mL)	0.27843
LOQ (µg/mL)	26.5175
Variance of calibration line	0.05153

In terms of LOD and LOQ, the sensitivity of the proposed analytical method was found as 0.278 and 26.518 µg/mL respectively (Table 1). The excipient's inclusion in the formulation had no effect on the drug peak suggesting that the approach is both specific and selective for the medication. The %recovery for the standard analysis and reference and analysis for all the three concentration levels (0.3, 0.4 and 0.5 µg/mL) ranged from 99.467 to 99.833% with confidence interval ranging from 0.001 to 0.002 at 95% confidence level for standard aloe-emodin solutions and % recovery 100.074 to 100.083, confidence interval 0.001 to 0.002 for concentration 0.3 µg/mL; % recovery 99.867 to 100.074 and confidence interval 0.002 to 0.002 for concentration 0.4 µg/mL and % recovery 99.697 to 100.056 with confidence interval 0.001 to 0.234, when spiked with 0.05, 0.10 and 0.15 µg/mL (Table 2). The results of the conventional addition and reference analysis approach were also discovered to be in agreement with the proposed method's correctness.²⁸

Table 2: Statistical data analysis for Accuracy/recovery of the pure drug

Conc. of test sample (µg/mL) (n=5)		Average ±SD	%RSD	% Recovery	Avg ±SD	CL at 95%	Minimum	Maximum	SAE
Taken	Added								
0.3	0	0.296 (0.001)	0.283	98.778	98.511 (0.279)	0.001	0.294	0.296	0.0004
0.4	0	0.401 (0.001)	0.325	99.833	100.283 (0.326)	0.002	0.399	0.402	0.0006
0.5	0	0.497 (0.001)	0.180	99.467	99.347 (0.179)	0.001	0.495	0.497	0.0004
0.3	0.05	0.350 (0.001)	0.239	100.381	100.038 (0.239)	0.001	0.349	0.350	0.0003
0.3	0.1	0.400 (0.001)	0.285	100.083	99.983 (0.285)	0.002	0.398	0.401	0.0006
0.3	0.15	0.450 (0.001)	0.254	100.074	99.941 (0.253)	0.002	0.448	0.451	0.0006
0.4	0.05	0.450 (0.001)	0.186	100.074	100.030 (0.186)	0.002	0.449	0.451	0.0005
0.4	0.1	0.499 (0.001)	0.229	99.867	99.787 (0.228)	0.002	0.497	0.500	0.0006
0.4	0.15	0.550 (0.001)	0.152	99.879	99.915 (0.152)	0.002	0.548	0.550	0.0005
0.5	0.05	0.549 (0.001)	0.152	99.697	99.842 (0.152)	0.001	0.548	0.550	0.0004
0.5	0.1	0.600 (0.002)	0.253	100.056	99.989 (0.253)	0.003	0.598	0.601	0.0009
0.5	0.15	0.650 (0.001)	0.129	100.051	100.021 (0.129)	0.234	99.897	100.205	0.0736

The intraday precision study data showed Standard Deviation in between 0.296±0.001 to 0.497±0.001 with %RSD 0.180 to 0.283, %recovery 98.511±0.279 to 100.283±0.326, Confidence Interval 0.001 to 0.002 and standard analytical error (SAE) 0.0004 to 0.0006 for the intraday assay precision for the pure aloe-emodin while for formulation 1 Standard Deviation in between 0.296±0.002 to 0.500±0.002 with %RSD 0.469 to 0.792, %recovery 98.644±0.506 to 100.067±0.469, Confidence Interval 0.002 to 0.004 and standard analytical error (SAE) 0.001 to 0.001 and for formulation 2, Standard Deviation in between 0.286±0.002 to 0.490±0.002 with %RSD 0.478 to 0.812, %recovery 95.311±0.505 to 98.067±0.469, Confidence Interval 0.002 to 0.004 and standard analytical error (SAE) 0.001 to 0.001. Whereas, inter day precision study data showed Standard Deviation in between 0.296 ±0.001 to 0.497±0.001 with %RSD 0.180 to 0.325, %recovery 98.511±0.279 to 100.283±0.326, Confidence Interval 0.001 to 0.002 and standard analytical error (SAE) 0.0004 to 0.0006 for the

pure aloe-emodin, while for formulation 1, Standard Deviation in between 0.300±0.003 to 0.501±0.001 with % RSD 0.228 to 0.983, %recovery 100.044±0.983 to 100.147±0.228, Confidence Interval 0.001 to 0.004 and standard analytical error (SAE) 0.0005 to 0.0013 and for formulation 2, Standard Deviation in between 0.290±0.001 to 0.490±0.002 with % RSD 0.449 to 0.812, % recovery 96.711±0.435 to 98.067±0.469, Confidence Interval 0.002 to 0.004 and standard analytical error (SAE) 0.002 to 0.004 respectively (Table 3). The results demonstrated the suggested good reproducibility, moderate precision, and intraday precision.

The variation in the maximum recovery within limitations of ±1.0 nm brought % recovery lying in the range of 98.178.279 to 98.844.279 with a maximum % confidence interval of 0.002 (Table 4) indicating the method sufficiently robust.²⁸

Table 3: Statistical data analysis of intraday and inter day precision for pure drug and drug in formulation

Test sample	Proposed methods	Amount (µg/mL)		RSD (%)	REC. (%) (±SD)	C.L.	Min	Max	SAE
		Taken	Found (±SD)						
Pure drug	Intraday assay (n=5)	0.3	0.296 (0.001)	0.283	98.511 (0.279)	0.001	0.294	0.296	0.0004
		0.4	0.401 (0.001)	0.325	100.283 (0.326)	0.002	0.399	0.402	0.0006
		0.5	0.497 (0.001)	0.180	99.347 (0.179)	0.001	0.495	0.497	0.0004
	Interday assay (n=5)	0.3	0.296 (0.001)	0.283	98.511 (0.279)	0.001	0.294	0.296	0.0004
		0.4	0.401 (0.001)	0.325	100.283 (0.326)	0.002	0.399	0.402	0.0006
		0.5	0.497 (0.001)	0.180	99.347 (0.179)	0.001	0.495	0.497	0.0004
Drug in formulation 1	Intraday assay (n=5)	0.3	0.296 (0.002)	0.512	98.644 (0.506)	0.002	0.294	0.298	0.0007
		0.4	0.399 (0.003)	0.792	99.833 (0.791)	0.004	0.394	0.402	0.0014
		0.5	0.500 (0.002)	0.469	100.067 (0.469)	0.003	0.496	0.502	0.0011
	Interday assay (n=5)	0.3	0.300 (0.003)	0.983	100.044 (0.983)	0.004	0.295	0.302	0.0013
		0.4	0.400 (0.002)	0.394	100.083 (0.395)	0.002	0.398	0.402	0.0007
		0.5	0.501 (0.001)	0.228	100.147 (0.228)	0.001	0.499	0.502	0.0005
Drug in formulation 2	Intraday assay (n=5)	0.3	0.286 (0.002)	0.530	95.311 (0.506)	0.002	0.284	0.288	0.0007
		0.4	0.389 (0.003)	0.812	97.333 (0.791)	0.004	0.384	0.392	0.0014
		0.5	0.490 (0.002)	0.478	98.067 (0.469)	0.003	0.486	0.492	0.0011
	Interday assay (n=5)	0.3	0.290 (0.001)	0.449	96.711 (0.435)	0.002	0.288	0.291	0.0016
		0.4	0.389 (0.003)	0.812	97.333 (0.791)	0.004	0.384	0.392	0.0039
		0.5	0.490 (0.002)	0.478	98.067 (0.469)	0.003	0.486	0.492	0.0029

Table 4: Robustness analysis data for the proposed method

Test sample (n=5)	Wavelength	Amount (µg/mL)		RSD (%)	REC. (%) (±SD)	C.L.	Min	Max	SAE
		Taken	Found (±SD)						
Pure drug	428.5	0.3	0.296 (0.001)	0.283	98.511 (0.279)	0.001	0.294	0.296	0.0003
		0.4	0.401 (0.001)	0.325	100.283 (0.326)	0.002	0.399	0.402	0.0006
		0.5	0.497 (0.001)	0.180	99.347 (0.179)	0.001	0.495	0.497	0.0004
	427.5	0.3	0.297 (0.001)	0.282	98.844 (0.279)	0.001	0.295	0.297	0.0004
		0.4	0.402 (0.001)	0.324	100.533 (0.326)	0.002	0.400	0.403	0.0006
		0.5	0.498 (0.001)	0.180	99.547 (0.179)	0.001	0.496	0.498	0.0004
	429.5	0.3	0.295 (0.001)	0.284	98.178 (0.279)	0.001	0.293	0.295	0.0004
		0.4	0.400 (0.001)	0.326	100.033 (0.326)	0.002	0.398	0.401	0.0006
		0.5	0.496 (0.001)	0.180	99.147 (0.179)	0.001	0.494	0.496	0.0004

Table 5: Assay data of pure drug and drug in marketed capsule formulations by proposed method

Sr. No.	Brand name	Formulation type	Amount labeled (mg)	Amount found (mg)	±SD	% RSD	% Recovery
1	Aloe- emodin, Avra Synthesis Pvt. Ltd.	Pure drug	100	99.46	0.084	0.017	99.89
2	Aloe- emodin, dietary supplement, Botany	Powder Formulation	100	100.01	0.064	0.013	100
3	Alfvital, Aloe- emodin Capsule	Capsule formulation	100	100.96	0.14	0.028	100.19

In all the marketed formulations of aloe-emodin, drug assay values ranged from 99.98 to 102.19%, with a maximum percentage confidence interval of ± 0.11 (Table 5).

CONCLUSION:

The analytical procedure created and validated on the UV1800 double beam spectrophotometer was straightforward, dependable, precise, and repeatable. The technique does away with the extraction phase which cuts down on analytical time, expense, and extraction errors. advantages are its cheap cost, fast speed, fair precision, and strong specificity, all of which enable a clear evaluation of the analyte in the presence of potentially present components. With no interference from excipients, the approach can now be conveniently utilized for routine quality control analysis of aloe-emodin in bulk pharmaceuticals, marketed tablets, and other formulations. It was successfully verified in compliance with the ICH standards. The method may be completely interchangeable with those already employed in plasma-based medication analysis; nevertheless, additional testing was required for the aforementioned disease.

CONFLICT OF INTEREST:

The authors report no financial or any other conflicts of interest in this work.

AVAILABILITY OF DATA AND MATERIALS:

Yes.

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